

THE WATER CYCLE

Water on Earth is always moving. It travels between the oceans, the air, and the land in a loop called the water cycle. The water that falls as rain today may have been in a river last week, in a cloud yesterday, or in an ocean long before that. No water is made and none is lost. The same water is used again and again, powered by heat from the sun and pulled down by gravity.

This reading follows one drop of water around the loop. It explains each step in plain words, gives an everyday example, and finishes with why the cycle matters to the people, plants, and animals that depend on it.

WHERE WATER IS FOUND

Most of Earth's water, about 97 percent, is salt water in the oceans. Only about 3 percent is fresh water, and most of that is locked away as ice in glaciers and at the poles. The small amount left is in lakes, rivers, soil, underground, and in the air. Even though the share in the air is tiny, it is the part that makes weather and brings rain, so it is very important.

Water exists in three forms. As a liquid it flows, as in a river or a glass of water. As a solid it is ice, snow, or hail. As a gas it is called water vapor, which is invisible and mixes with the air. The water cycle is mostly the story of water changing between these three forms.

EVAPORATION

The sun warms the water in oceans, lakes, and rivers. As the water heats up, its tiny particles move faster. Some of them move so fast that they escape from the surface and rise into the air as a gas called water vapor. This change from liquid to gas is called evaporation.

Evaporation happens faster when it is hot, dry, or windy. A shallow puddle on a sunny afternoon can disappear in an hour, while the same puddle on a cool, damp day may stay for days. The oceans supply most of the water vapor in the air, because they cover most of the planet.

An everyday example is wet laundry on a clothesline. The clothes dry because the water in them evaporates into the air. Another is the steam that rises from a cup of hot tea. The steam you can see is already turning back into tiny drops, but the invisible vapor right above the cup came from evaporation.

Salt and other minerals do not evaporate with the water. When seawater evaporates, the salt is left behind. This is why rain is fresh water even though much of it began in the salty ocean.

TRANSPIRATION

Plants also send water into the air. A plant takes up water through its roots and carries it up through its stem to its leaves. The plant keeps only a small part of this water. Most of it leaves through tiny openings in the leaves and joins the air as water vapor. This is called transpiration.

Transpiration helps the plant in two ways. It pulls more water and nutrients up from the soil, and it cools the leaves, much as sweating cools a person. A large tree can give off hundreds of liters of water on a warm day, and a whole forest puts a great deal of moisture into the air. This is one reason forests are often damp and cool, and why cutting down large areas of forest can change the local weather.

Scientists often put evaporation and transpiration together and call the combined total evapotranspiration, because from far above it is hard to tell which one a drop of vapor came from.

CONDENSATION

Water vapor rises into the sky, where the air is cold. As the vapor cools, its particles slow down and cling together. The vapor turns back into tiny drops of liquid water. This change from gas to liquid is called condensation.

The drops are so small and light that they float. Billions of them gather around specks of dust and salt in the air, and together they form a cloud. A cloud is not water vapor. It is a mass of tiny liquid drops, or in very cold air, tiny ice crystals.

Condensation also happens close to the ground. The drops that form on the outside of a cold glass on a warm day are water vapor from the room air condensing on the cold surface. Morning dew on grass and the fog over a lake at dawn form in the same way, when the air near the surface cools enough that its vapor turns to liquid.

PRECIPITATION

Inside a cloud, the tiny drops bump into each other and join together, becoming larger and heavier. When the drops grow too heavy for the air to hold up, they

fall. Any water that falls from clouds to the ground is called precipitation.

What falls depends on the temperature of the air. When it is warm enough all the way down, the drops fall as rain. When the air is below freezing, the water freezes into ice crystals and falls as snow. Sleet forms when raindrops freeze as they fall through a layer of cold air. Hail forms inside strong storm clouds, where wind carries drops up and down through freezing air, and each trip adds another layer of ice until the stone is heavy enough to fall.

The amount of precipitation is very different from place to place. Some deserts get less than 25 millimeters of rain in a year, while some tropical mountains receive more than 10 meters.

COLLECTION

Water that falls does not disappear. Some of it runs over the surface of the land and gathers in streams, which join to form rivers, which flow to lakes and finally to the ocean. This water is called runoff. Some of the water soaks into the ground. It moves slowly through soil and rock and is stored underground as groundwater. In many places people reach groundwater by digging wells.

Some of the precipitation falls as snow in cold places and stays frozen for months or even thousands of years in glaciers and ice sheets. When the snow melts in spring, it feeds rivers with fresh water. Much of the water that people drink comes from rivers, lakes, and groundwater that began as precipitation.

Plants take up some of the water that soaks in. The sun warms the water in the oceans, lakes, and rivers again, and more of it evaporates. The cycle begins once more. It has no start and no end.

PUTTING IT TOGETHER

Here is the whole loop in order. The sun heats water, and some of it evaporates. Plants add to this when they transpire. The vapor rises and cools, and condenses into clouds. The drops in the clouds grow heavy and fall as precipitation. The water is collected in rivers, lakes, oceans, and underground, and the cycle repeats.

The time one drop spends in each place varies enormously. A drop may stay in the atmosphere for only about nine days, in a river for a couple of weeks, in a lake for years, and deep in the ocean or in an ice sheet for thousands of years.

WHY THE WATER CYCLE MATTERS

The water cycle moves fresh water to the land. Without it, rain would not fall on farms and forests, and rivers would dry up. It also spreads heat around the planet. When water evaporates it takes heat from the surface, and when it condenses high in the sky it gives that heat back to the air, which helps drive winds and storms.

People can affect the cycle. Cities cover soil with roads and roofs, so rain runs off quickly instead of soaking in. Cutting forests reduces transpiration, which can mean less rain nearby. Pollution on the land can be carried by runoff into rivers and groundwater. Protecting wetlands, planting trees, and using water carefully all help the cycle keep working for everyone.

KEY TERMS

Water cycle: the continuous movement of water between the oceans, the air, and the land.

Evaporation: liquid water changing into a gas called water vapor.

Transpiration: water leaving a plant through its leaves as vapor.

Condensation: water vapor cooling and changing into liquid droplets.

Precipitation: water falling from clouds to the ground as rain, snow, sleet, or hail.

Collection: fallen water gathering in rivers, lakes, oceans, and underground.

Runoff: water that flows over the land into streams and rivers.

Groundwater: water that has soaked into the soil and rock underground.